
CAPSTONE PROJECT

COLLEGE ADMISSION ASSISTANT USING RAG-BASED AGENT ON IBM WATSONX.AI

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OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

Manual college admission processes involve time-consuming inquiries about eligibility, courses, fees, cutoffs, and deadlines. Students often struggle to get accurate, updated information from official sources or helplines . There is a growing need for an AI-powered assistant that can provide transparent, real-time, and personalized responses to student queries related to admissions using trusted institutional data.

PROPOSED SOLUTION

- ❖ The proposed system is a College Admission Agent built using IBM Watsonx.ai and powered by RAG (Retrieval-Augmented Generation). It helps students get accurate admission details through natural language queries.
- ❖ **Key components:**
 - Retrieval: Uses Tavily and Wikipedia Search to fetch real-time data
 - Generation: Uses Granite-13B Chat Model to generate human-like responses
 - Built on IBM Cloud Lite using Watsonx.ai Studio
 - Deployed in Watsonx Deployment Space and tested with real student queries
- ❖ This AI-powered agent simplifies the admission process, reduces manual queries, and improves access to reliable, up-to-date information.

SYSTEM APPROACH

❖ Technologies Used:

- IBM Cloud Lite
- Watsonx.ai Studio
- Granite-13b-chat Foundation Model
- Tavily and Wikipedia Search Tools (retrievers)
- Agentic AI Builder for RAG setup

❖ Steps:

- Create Watsonx.ai project
- Build Agentic AI interface
- Configure tools (Tavily, Wikipedia)
- Define prompts
- Deploy in Deployment Space

ALGORITHM & DEPLOYMENT

❖ RAG Architecture:

- Retrieval: Tavily/Wikipedia fetch real-time info
- Augmentation: Retrieved content injected into the prompt
- Generation: Granite model crafts a relevant response

❖ Deployment:

Promoted to IBM Watsonx Deployment Space

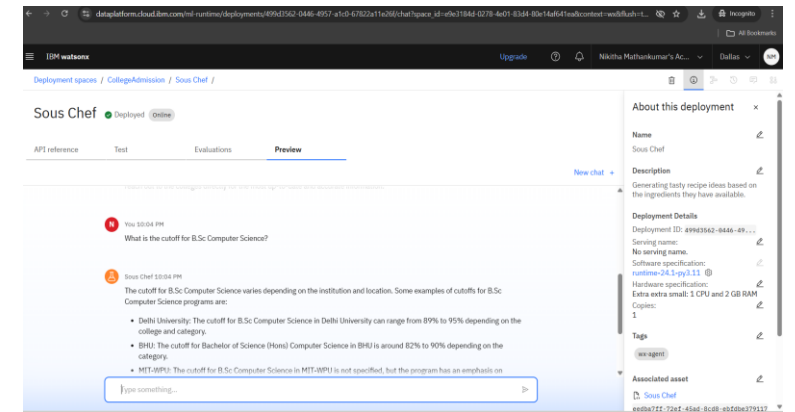
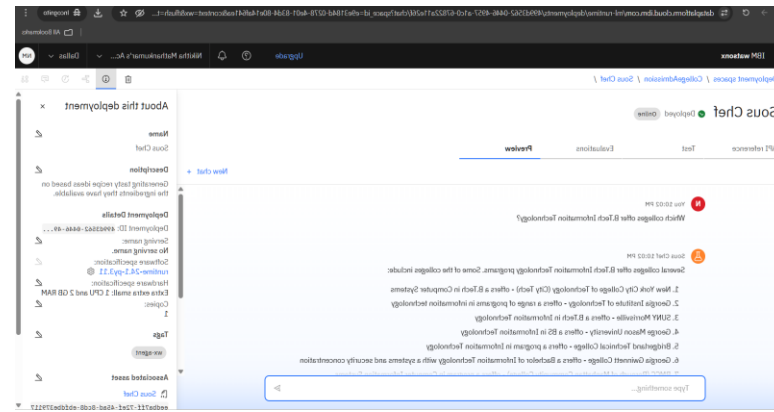
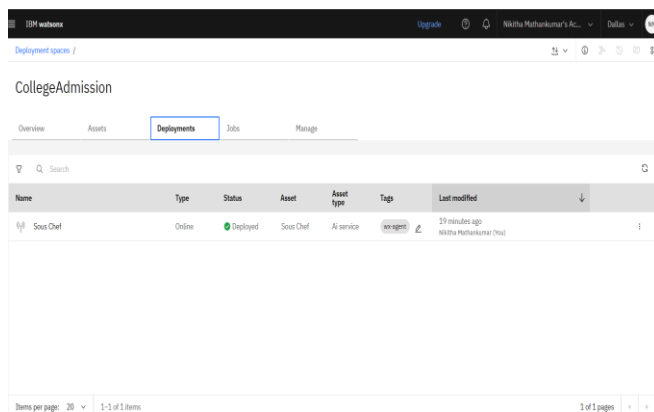
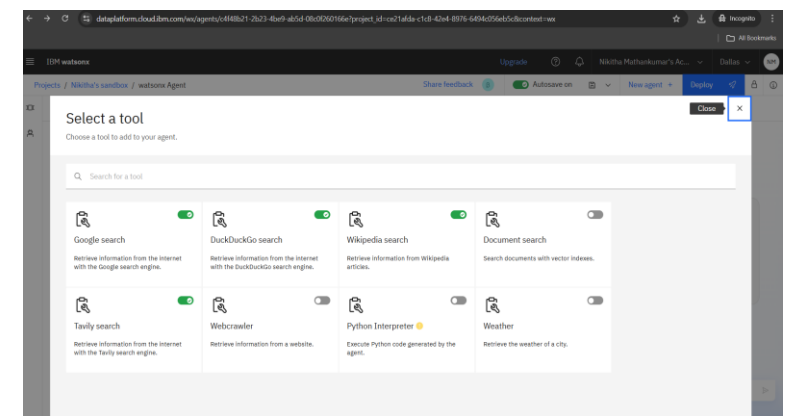
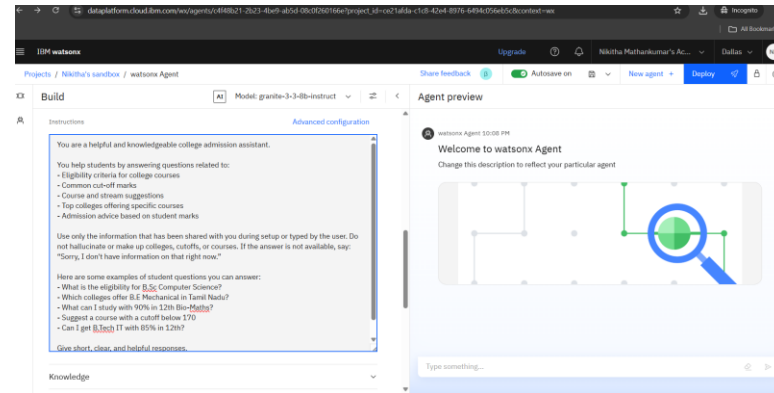
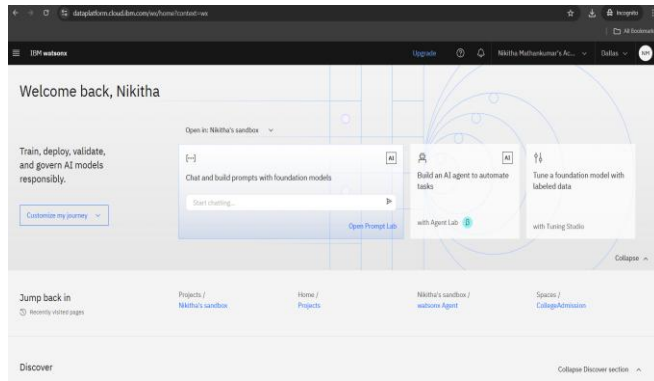
Live tested through Watsonx Preview and API

RESULT

Example User Queries:

- What is the eligibility for B.E Computer Science and Engineering?
- Which colleges offer B.Tech Information Technology?
- What is the cutoff for B.Sc Computer Science?
- I have 180 cutoff. What can I apply for?
- Suggest courses that accept 160 cutoff.

RESULT



CONCLUSION

The College Admission Agent developed using IBM Watsonx.ai successfully demonstrates the power of Retrieval-Augmented Generation (RAG) in simplifying the student admission process. By combining the Granite-13B foundation model with real-time data retrieval through tools like Tavily and Wikipedia, the agent provides accurate, conversational answers to common admission queries. This AI-driven solution enhances accessibility, reduces manual effort, and ensures that students receive timely and reliable guidance on eligibility, courses, cutoffs, and colleges — all through a simple, user-friendly interface hosted entirely on IBM Cloud.

FUTURE SCOPE

- Add Document Search to use internal PDFs (cutoff list, brochures)
- Extend to support application tracking, document uploads
- Support multiple languages for regional access
- Connect to official APIs from university databases

REFERENCES

- [1] L. Nguyen and T. Quan, “URAG: Implementing a Unified Hybrid RAG for Precise Answers in University Admission Chatbots — A Case Study at HCMUT,” arXiv preprint arXiv:2501.16276, Jan. 2025. [Online]. Available: <https://arxiv.org/abs/2501.16276>
- [2] A. Nguyen-Duc, C. V. Manh, B. A. Tran, V. P. Ngo, L. L. Chi, and A. Q. Nguyen, “An Empirical Study of Multi-Agent RAG for Real-World University Admissions Counseling,” arXiv preprint arXiv:2507.11272, Jul. 2025.

IBM CERTIFICATIONS



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Completion Certificate



This certificate is presented to

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**Lab: Retrieval Augmented Generation with
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According to the Adobe Learning Manager system of record

Completion date: 24 Jul 2025 (GMT)

Learning hours: 20 mins



THANK YOU